

Dipayan Biswas

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Summary

PhD candidate in Computer Science (expected Dec 2026) with 5+ years in multimodal AI and computer vision, building and evaluating vision-language model (VLM) systems end-to-end: dataset construction, fine-tuning, multi-backend inference, and LLM-as-judge evaluation. Shipped multimodal summarization to a video platform used by 750+ students across 10+ courses. 5 IEEE publications.

Education

University of Houston

Ph.D. in Computer Science, GPA 3.88 (M.Sc. in Computer Science awarded May 2026)

Houston, TX
Aug 2021 – Dec 2026 (Expected)

Khulna University of Engineering & Technology (KUET)

B.Sc. in Electronics and Communication Engineering, GPA 3.88, University Gold Medalist

Khulna, Bangladesh
Apr 2012 – May 2016

Technical Skills

Specializations: Deep Learning, Multimodal AI, Vision-Language Models (VLMs), LLMs, Computer Vision, Object Detection, Cross-modal Fusion, LLM Evaluation (LLM-as-judge), RAG, In-context Learning, Fine-tuning (QLoRA), Semi-supervised Learning

Frameworks: PyTorch, HuggingFace Transformers, TensorFlow, OpenCV, LangChain, scikit-learn, DeepEval, Hydra

Models & APIs: Gemini API, OpenAI API, Gemma 3, Qwen3-VL, Whisper, CLIP, SigLIP, YOLO11, DETR

MLOps & Languages: Python, C++, SQL, Docker, Git, FastAPI, vLLM, Ollama, MLflow, Slurm, AWS (SageMaker, EC2, S3)

Experience

University of Houston

Graduate Research / Teaching Assistant

Houston, TX
Aug 2021 – Present

- Led multimodal research on the NSF-funded VideoPoints platform, distilling hour-long lectures into hierarchically structured summaries with supporting visuals, deployed across 10+ university courses serving 750+ students.
- Designed a cross-modal fusion pipeline over ASR transcripts, OCR text, and video frames via VLMs and in-context learning, processing 25k hours of lecture video into chapter-level summaries with auto-selected supporting visuals.
- Built an automated LLM-as-judge evaluation harness (DeepEval, Gemini/OpenAI APIs) over 500 lecture summaries, raising factual faithfulness from 86.4% to 96% and relevance from 78.8% to 84% versus a transcript-only baseline.
- Raised slide-element detection from 90.8% to 95.3% AP50 by combining YOLOv11 with semi-supervised auto-labeling, expanding the labeled set from 1k expert-labeled to 4k frames with no added annotation effort (IEEE MIPR 2025).
- Developed a graph-based method that groups fragmented detections into whole slide objects (+15.8% mAP, IEEE ISM 2023).

Fellowship.AI

Machine Learning Fellow, Team Lead

Remote
Sep 2025 – Nov 2025

- Led a fellowship team building a multimodal listing analyzer (Gemini API) that jointly scores listing images and descriptions to produce structured pricing, staging, and content recommendations.
- Engineered a weighted-KNN price prediction pipeline with outlier detection over 8K+ Redfin listings to flag mispriced properties.

Khulna University of Engineering & Technology (KUET)

Assistant Professor (promoted from Lecturer)

Khulna, Bangladesh
Feb 2017 – Aug 2021

- Published deep-learning research on plant disease (99.80%) and breast cancer (99.28%) classification at IEEE EICT and ICAREE.
- Mentored 10+ undergraduates in ML research and taught five undergraduate courses, including Digital Image Processing.

Samsung R&D Institute Bangladesh (SRBD)

Software Engineer

Dhaka, Bangladesh
Jul 2016 – Feb 2017

- Implemented panoramic, 360-degree, and little-planet rendering modes (Objective-C, OpenCV, OpenGL) for the shipped Samsung Gear 360 iOS app, serving real-time playback of captured 360 video.

Selected Projects

Unified VLM Inference Framework [Code]

Jun 2026 – Aug 2026

- Engineered unified VLM inference across 5+ local and cloud backends (Ollama, vLLM, HuggingFace, OpenAI, Gemini) with a shared interface for interleaved text-image prompting and multi-step workflows, switchable with one config change.

Lecture Video Chaptering (LLM-Based Temporal Segmentation)

Jun 2025 – Aug 2025

- Built a chaptering pipeline aligning Whisper transcripts with slide transitions, validated for gap-free chapters; deployed in production.

Selected Publications

- Biswas, Shah, Subhlok.** "Visual Content Detection in Educational Videos with Transfer Learning and Dataset Enrichment." *IEEE MIPR 2025*. [Code]
- Biswas, Shah, Subhlok.** "Lecture Video Visual Objects (LVVO) Dataset: A Benchmark for Visual Object Detection." *arXiv 2025*. [Data]
- Biswas, Shah, Subhlok.** "Identification of Visual Objects in Lecture Videos with Color and Keypoints Analysis." *IEEE ISM 2023*.

Honors, Certifications & Leadership

- Honors:** Ph.D. Research Showcase Award (Audience's Choice), UH 2025 • Cullen Graduate Fellowship, UH 2021–2026 • Gold Medalist, KUET
- Certifications:** Generative AI with LLMs (AWS, DeepLearning.AI) • Building and Evaluating Advanced RAG • Machine Learning in Production